DM – Permutations and combinations

NGUYEN Hoang Thach
nhthach@math.ac.vn

Outline

1. Permutations
2. Combinations
3. Binomial theorem and identities

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A cats' popularity contest

Eight cats enter a popularity contest. The results will be determined by online voting. How many possible total rankings are there?

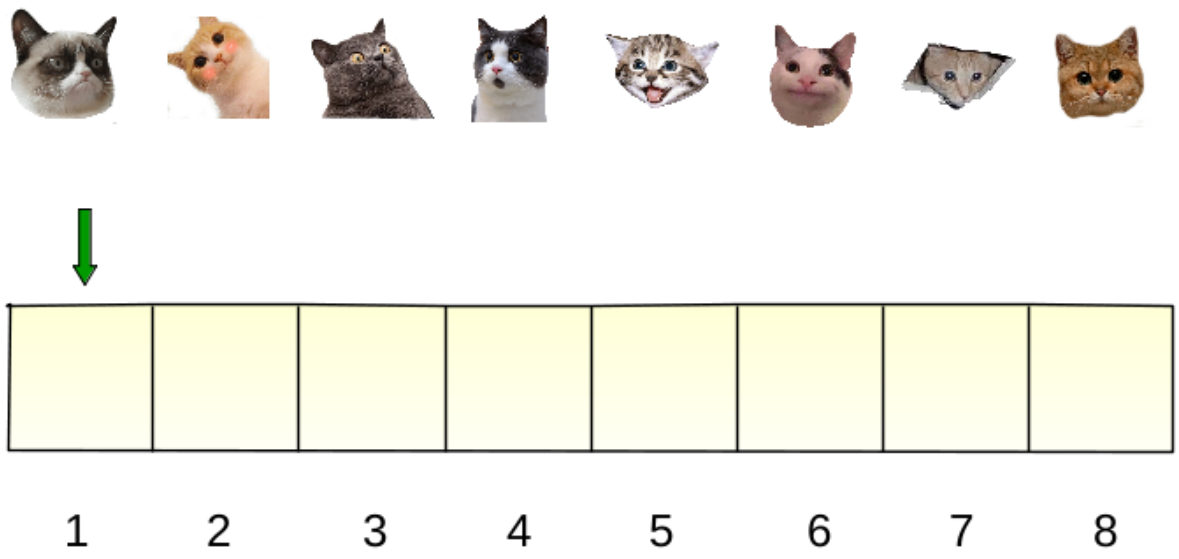


A cats' popularity contest



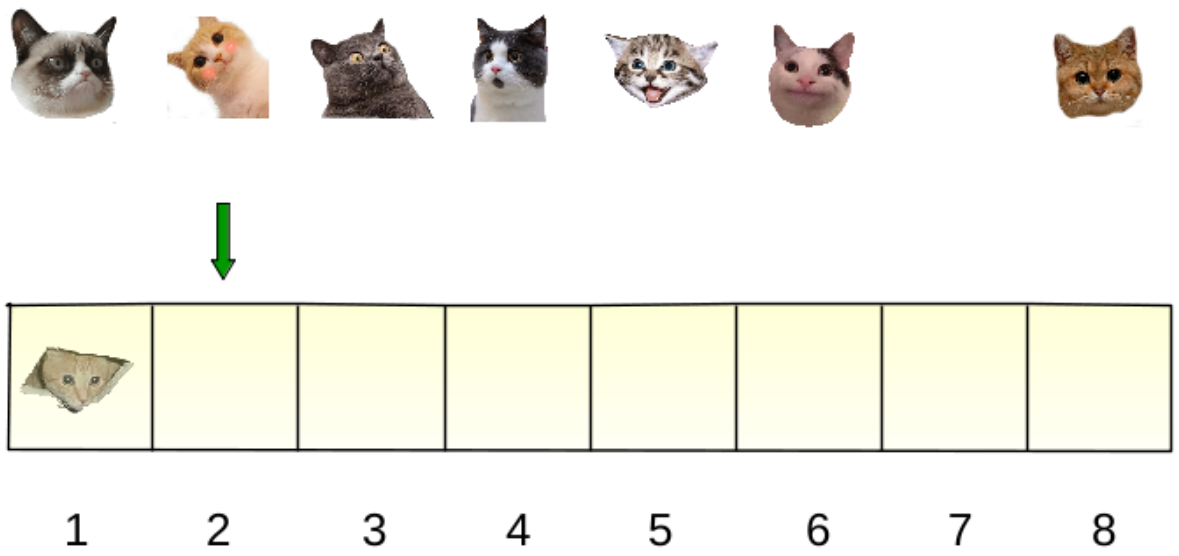
A total ranking is an *arrangement* of the cats in a *linear order*.

A cats' popularity contest



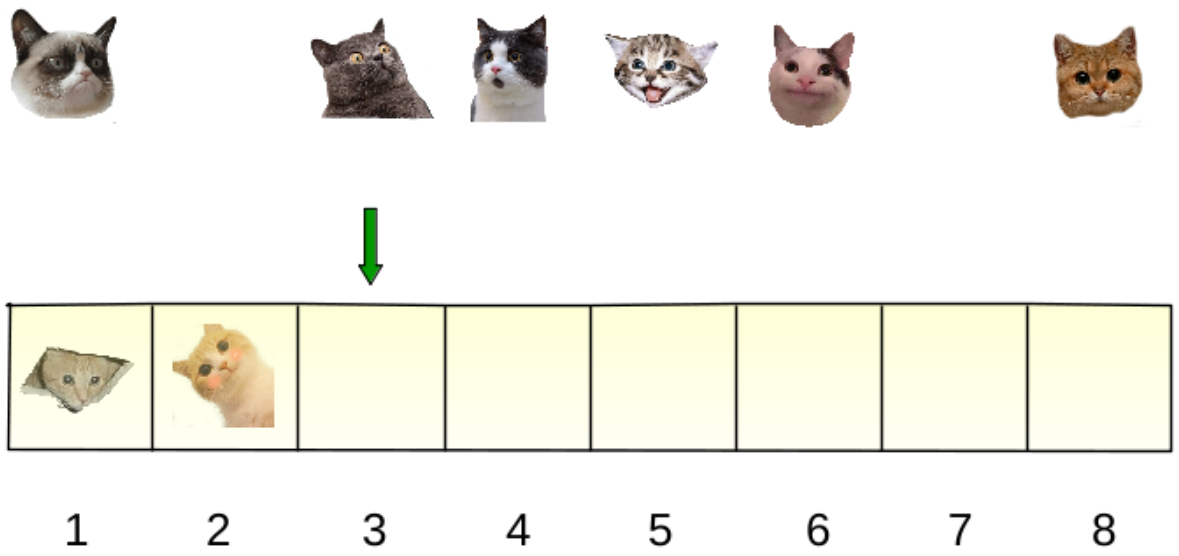
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Second place: 7 choices;
Third place: 6 choices;
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Eighth place: 1 choices;

A cats' popularity contest



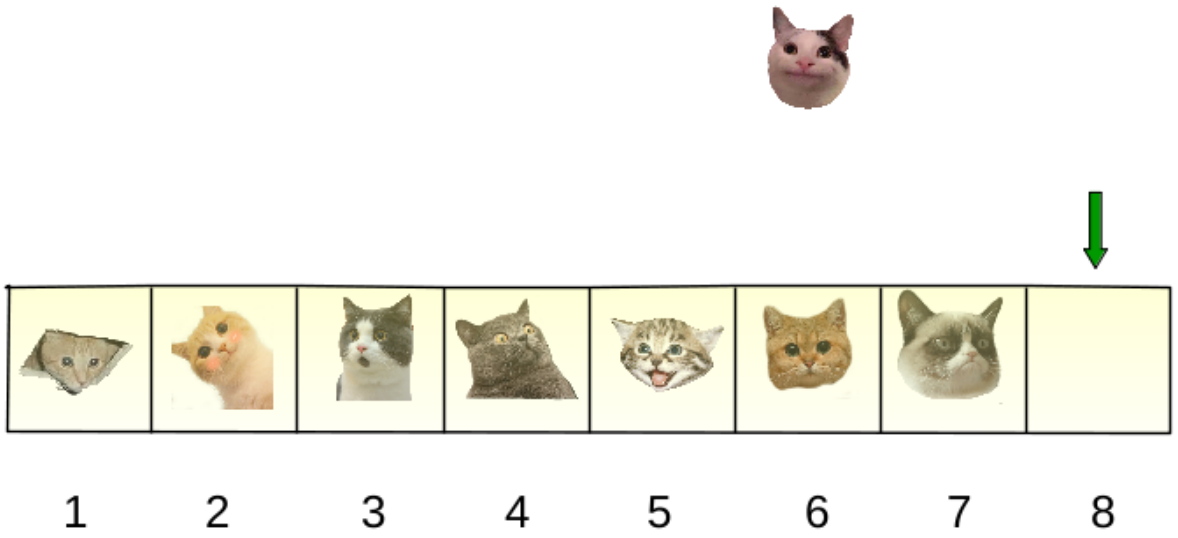
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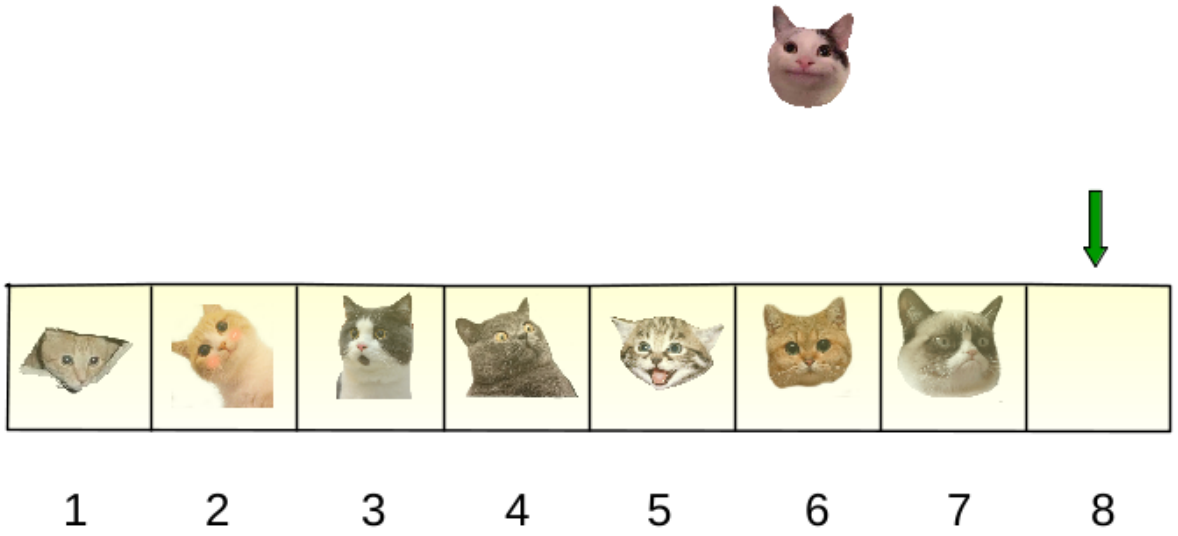
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A cats' popularity contest



First place: 8 choices;
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$$8 \times 7 \times 6 \times \cdots \times 1 = 40320 \text{ possible rankings}$$

Permutations

Definition

A permutation of a set is an ordered arrangement of the elements of that set.

Theorem

The number of permutations of a set of n elements is

$$n! := n \times (n-1) \times \cdots \times 2 \times 1\,.$$

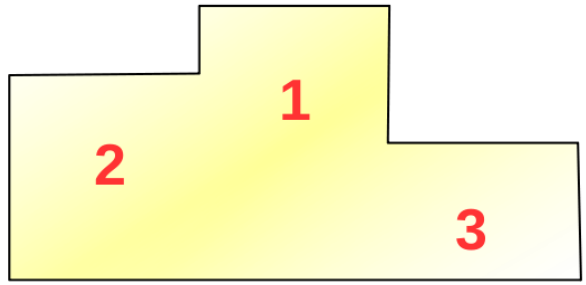
(Read: "n factorial".)

By convention, $0! = 1\,.$

Remark: $n! = n \times (n-1)!$ for all $n \geq 1$.

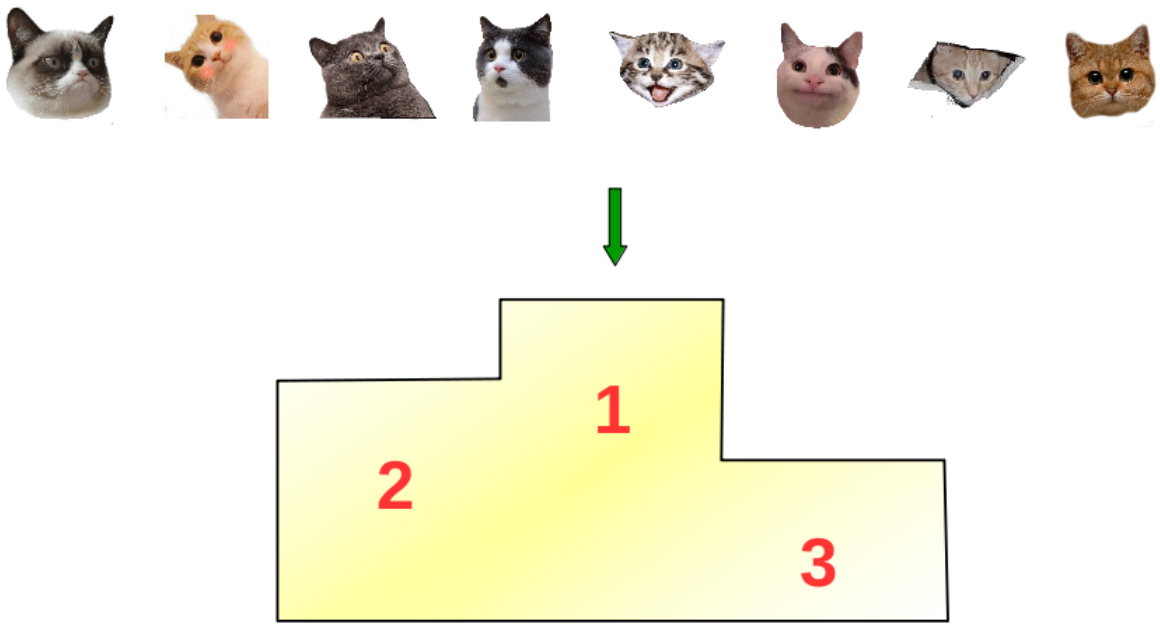
A cats' popularity contest (2)





How many possible top 3?

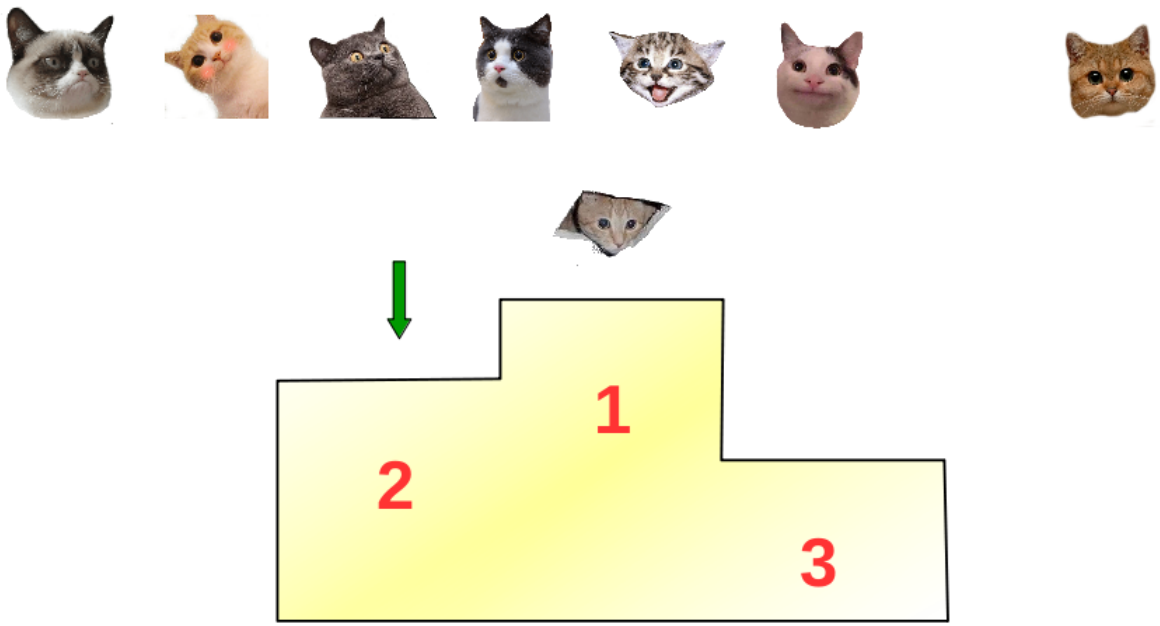
A cats' popularity contest (2)



First place: 8 choices;
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Third place: 6 choices;

$$8 \times 7 \times 6 = 336 \text{ possible top three}$$

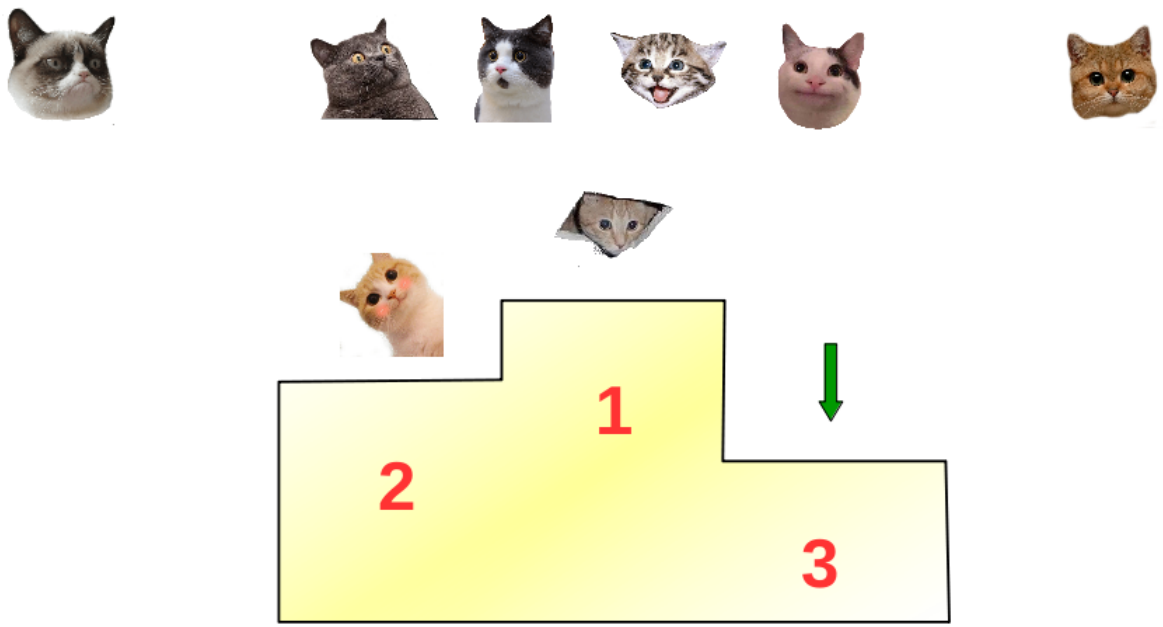
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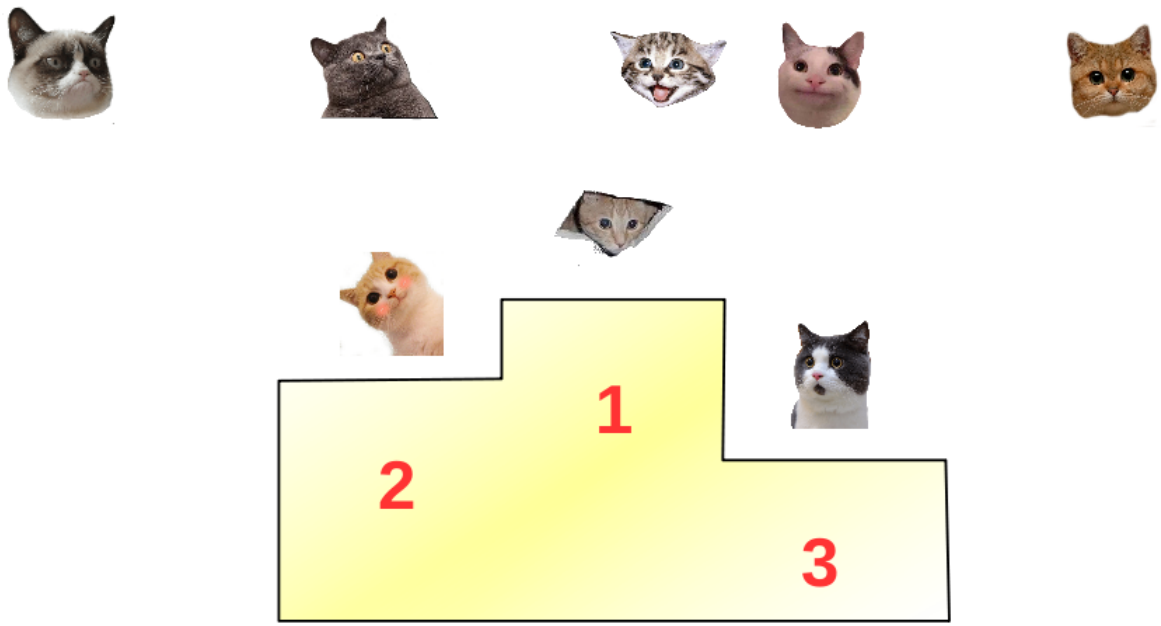
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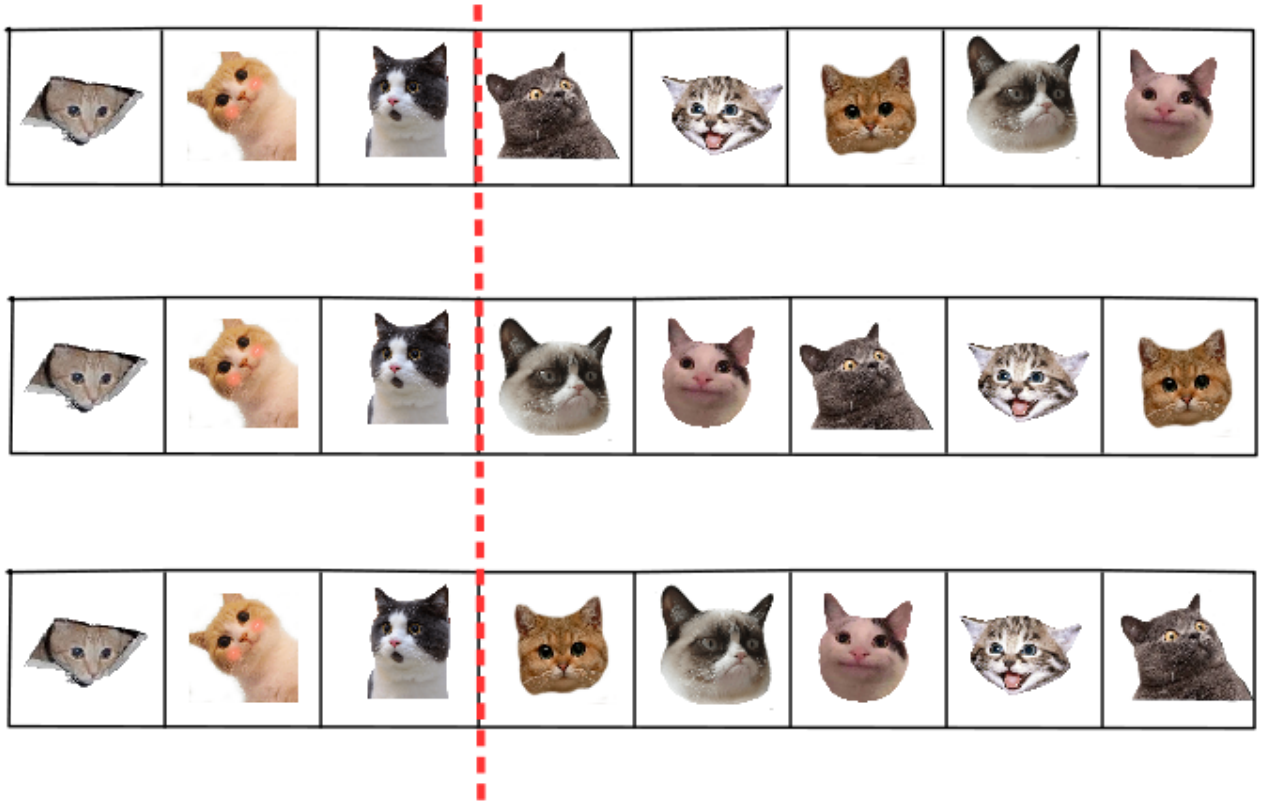
Top three and total ranking

One can take a ranking and extract the top three from it:



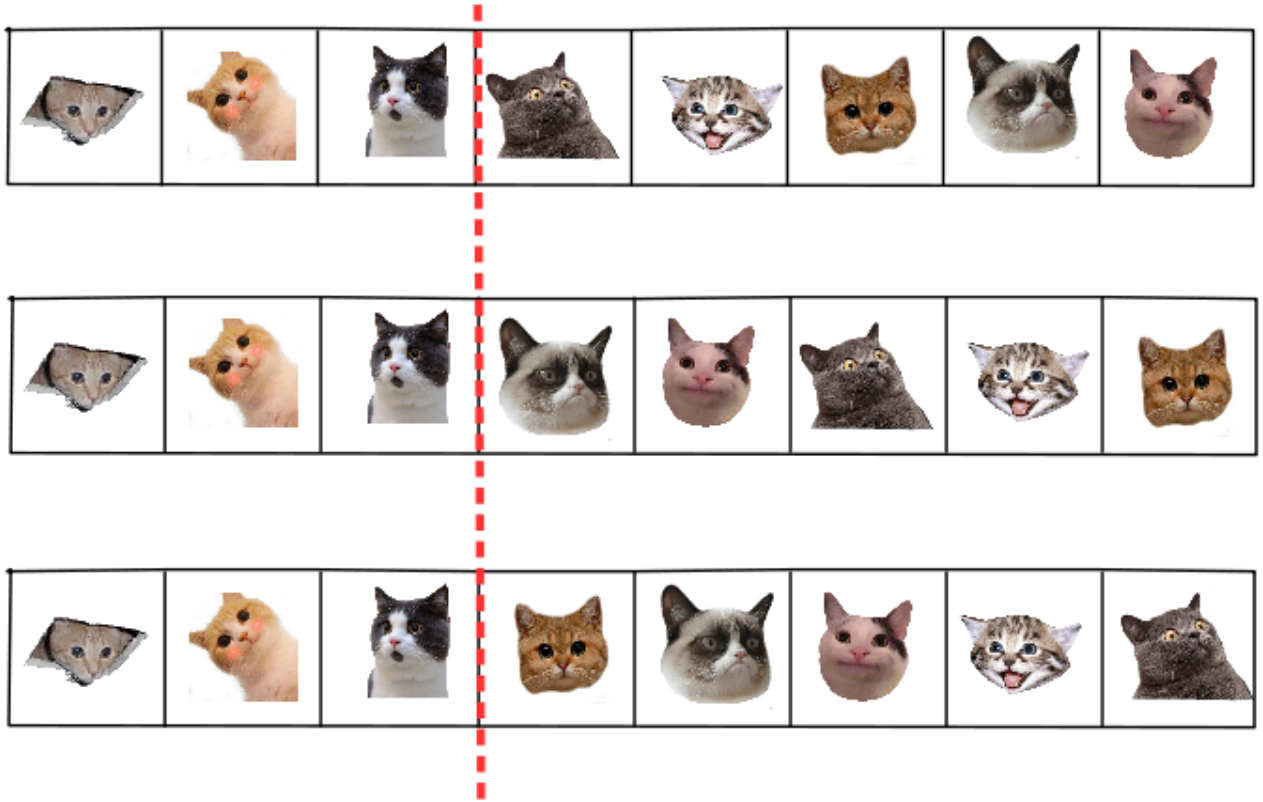
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But each top three can be obtained this way from *different* rankings:



Top three and total ranking

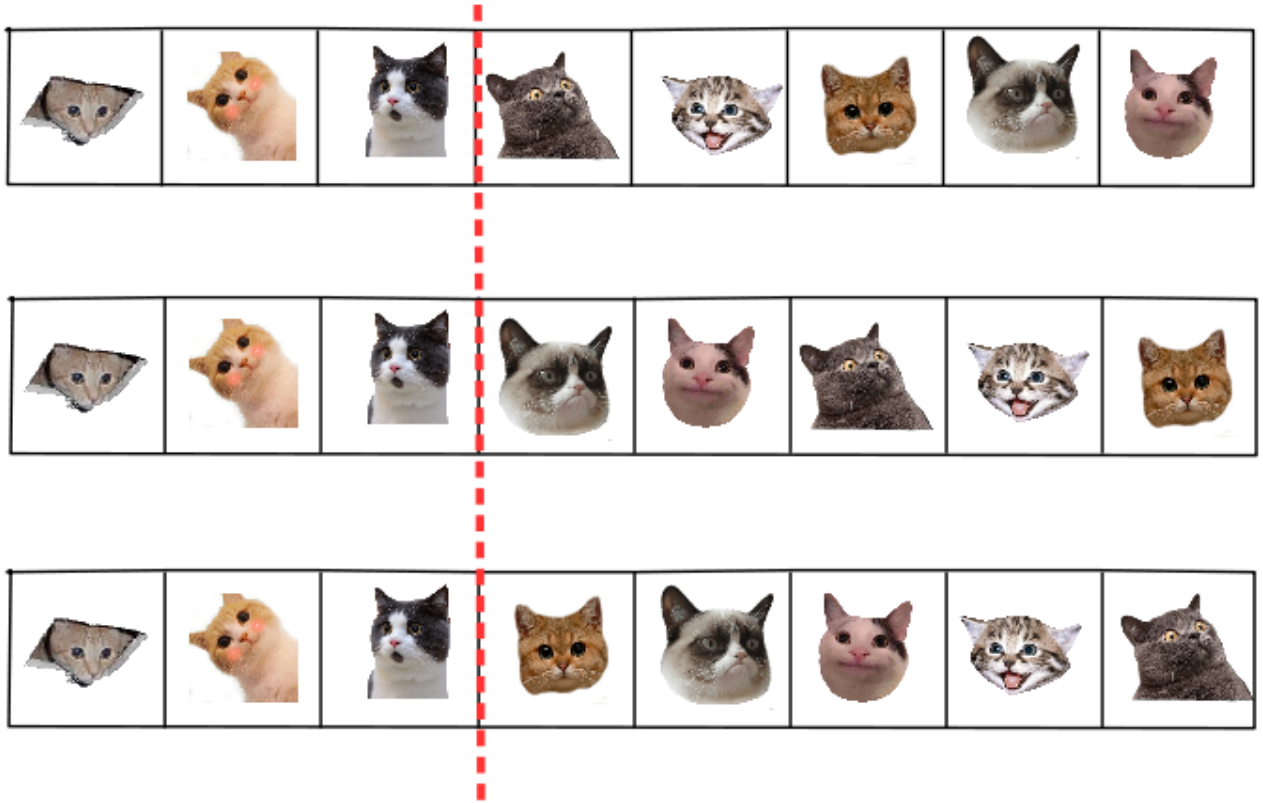
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Top three and total ranking

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Each top three can be obtained this way from exactly 5! rankings!

$$\#\{\text{top threes}\} = \frac{\#\{\text{rankings}\}}{5!} = \frac{8!}{5!}$$

r-permutations

Definition

An r-permutation of a set is an ordered arrangement of r elements of that set.

Theorem

The number of r-permutations of a set of n elements, where $1 \leq r \leq n$ is

$$P(n, r) := n \times (n-1) \times \cdots \times (n-r+1) = \frac{n!}{(n-r)!}\,.$$

By convention, $P(n, 0) = 1$.

Examples

A salesman has to present a product in eight different towns. He starts his trip in his home town, and from each town he can go to any other town. How many possible iteneraries are there?



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Since the first town is fixed and the other towns can be visited in any order, the number of possible visiting orders is the number of permutations of the seven other towns.

Ans: $7! = 5040$.

Examples

What is the probability that a random permutation of the string ABCDEFG contains the pattern ABC?

Consider ABC a single block. A permutation of $ABCDDEFG$ containing ABC is none other than a permutation of five elements: ABC, D, E, F, G. There are $5! = 120$ such strings.

Out of $7! = 5040$ permutations of $ABCDEFG$, there are 120 that contain ABC. Thus, the probability is $120/5040 \approx 2.38\%$.

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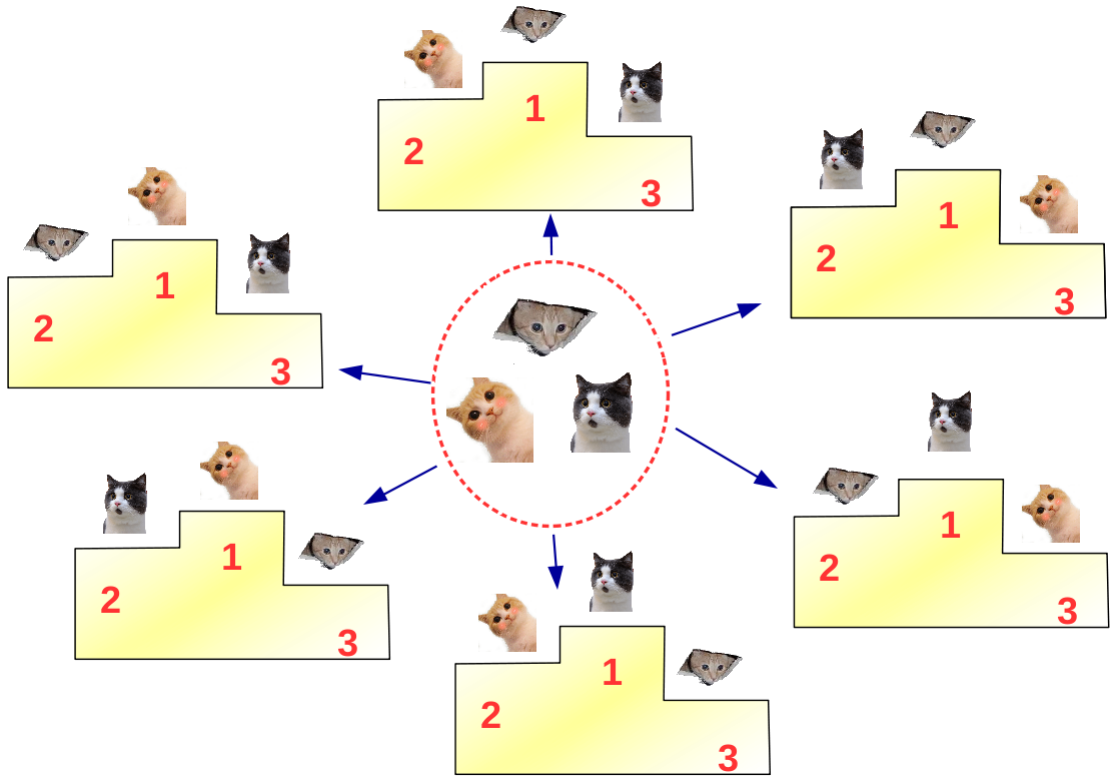
A cats' popularity contest (3)

Top three cats will represent the town at the regional contest. How many possible three-cat teams are there?



A cats' popularity contest (3)

Observation: Order among the top three is not important in this case.



Each team of three corresponds to $3! = 6$ different top threes.

$$\#\{\text{teams of three}\} = \frac{\#\{\text{top threes}\}}{3!} = \frac{336}{6} = 56\,.$$

Combinations

Definition

An r-combination of a set is an unordered selection of r elements from that set. In other words, an r-combination of a set is a subset of r elements of that set.

The number of r-combinations of a set of n elements is denoted by $C(n, r)$ *or* $\binom{n}{r}$ *[read: 'n choose r'].*

Number of combinations

Theorem

For non-negative intergers n and r such that $r \leq n$:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}\,.$$

Proof (by *double counting*)

Consider the following way of creating the r-permutations of n elements:

- **Step 1:** Choose r elements. There are $C(n, r)$ ways.
- **Step 2:** Arrange those r elements. There are $r!$ ways.

By the product rule, there are in total $C(n, r) \times r!$ ways.

Thus:

$$P(n, r) = C(n, r) \times r!\,.$$

which implies

$$C(n, r) = \frac{P(n, r)}{r!} = \frac{n!}{r!(n-r)!}\,.$$

Examples

How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

Ans:

- Hands of five: $n = 52, r = 5,\ C(52, 5) = 52!/(5!47!) = 2,598,960$ ways.
- Hands of 47: $n = 52, r = 47,\ C(52, 47) = 52!/(47!5!) = 2,598,960$ ways.

Corollary

If $0 \leq r \leq n$, *then* $C(n, r) = C(n, n - r)$.

Proof: Each subset of r elements corresponds to one and only one subset of $n - r$ elements.

Note: Proof by bijection (combinatorial proof).

Examples

Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

Ans: There are:

- $C(9,3)$ ways to select 3 maths representatives,
- $C(11,4)$ ways to select 4 CS representatives.

By the product rule, the number of possible committees is $C(9,3) \times C(11,4) = 27720$.

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Binomial theorem

Theorem

Let n be a non-negative integer. Then

$$
\begin{aligned}
(x+y)^n &= \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i \qquad (1)\\
&= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n .
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Examples:

$$(x+y)^1 = 1 \cdot x + 1 \cdot y$$
$$(x+y)^2 = 1 \cdot x^2 + 2xy + 1 \cdot y^2$$
$$(x+y)^5 = 1 \cdot x^5 + 5x^4 y + 10x^2 y^3 + 10x^3 y^2 + 5xy^4 + 1 \cdot y^5$$

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Proof

For each i, the number of $x^{n-i}y^i$ terms in the expansion of the LHS is equal to the number of ways to choose $n-i$ x's and i y's from the sums $(x+y)$. There are $\binom{n}{n-i} = \binom{n}{i}$ ways to choose, hence the coefficient of $x^{n-i}y^i$ in the expansion is $\binom{n}{i}$.

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Proof

Example with $n = 3, i = 2, n - i = 1$:

$$\begin{aligned}(\mathbf{x}+y)\,(x+\mathbf{y})\,(x+\mathbf{y}) &\longrightarrow xyy = xy^2\\ (x+\mathbf{y})\,(\mathbf{x}+y)\,(x+\mathbf{y}) &\longrightarrow yxy = xy^2\\ (x+\mathbf{y})\,(x+\mathbf{y})\,(\mathbf{x}+y) &\longrightarrow yyx = xy^2\end{aligned}$$

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(x + \mathbf{y})\,(x + \mathbf{y})\,(\mathbf{x} + y) &\longrightarrow yyx = xy^2
\end{aligned}
$$

Note: The formula can also be proven by induction on n.

Examples

- What is the coefficient of x^6y^2 in the expansion of $(x+y)^8$?
  Ans: $\binom{8}{2} = 28$.

- What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$.
  Ans: $\binom{25}{12}2^{12}(-3)^{13} = -\frac{25!}{12!13!}2^{12}3^{13}$.

Corollaries

Let $x = y = 1$ in (1). Then:

Corollary

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n .$$

Note: This identity can also be proven by counting the number of subsets of $\{1, 2, \ldots, n\}$ in two ways.

Let $x = -1, y = 1$ in (1). Then:

Corollary

$$\sum_{i=1}^{n} (-1)^i \binom{n}{i} = 0 .$$

Pascal's identity

Theorem

Let n and k be positive integers such that $n \geq k$. Then:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}. \tag{2}$$

Proof

Let $S = \{1, 2, \ldots, n+1\}$. The LHS counts the number of subsets of size k of S.

We recount this quantity in a different way. The subsets of size k of S belong to either of the two types: those that contain $n+1$ and those that do not. There are $\binom{n}{k-1}$ subsets of the first type and $\binom{n}{k}$ subsets of the second type.

Pascal's triangle

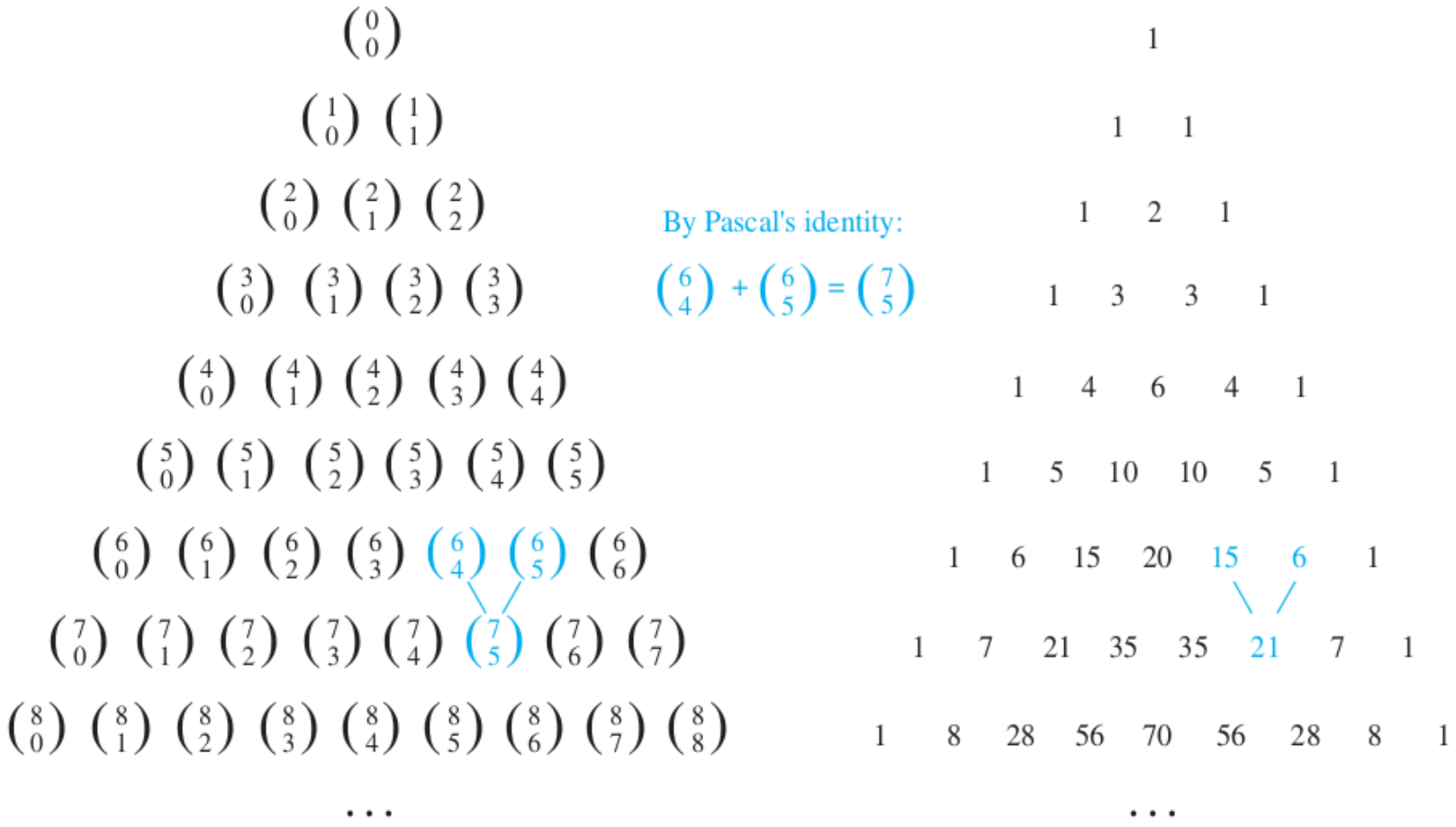


Figure: Pascal's triangle (Rosen, p. 419)

Vandermonde's identity

Theorem

Let m, n, r be non-negative integers such that $r \leq m$ and $r \leq n$. Then:

$$\binom{m+n}{r} = \sum_{i=0}^{r} \binom{m}{r-i}\binom{n}{i}. \tag{3}$$

Proof

The LHS counts the number of subsets of size r of the set $\{1, 2, \ldots, m+n\}$.

Such a subset can be formed by choosing i elements from $\{1, 2, \ldots, n\}$ first, then $r - i$ elements from $\{n+1, n+2, \ldots, n+m\}$. The second way of counting yields the RHS.

Vandermonde's identity

Let $r = m = n$ in (3), we obtain:

Corollary

$$\binom{2n}{n} = \sum_{i=0}^{n} \binom{n}{i}^2.$$

Counting bit string

Theorem

Let n, r be non-negative integers such that $r \leq n$. Then:

$$\binom{n+1}{r+1} = \sum_{i=r}^{n} \binom{i}{r}. \tag{4}$$

Proof

The LHS counts the number of bit strings of length $n+1$ having $r+1$ 1's. Such a string can be formed by adding 0's at the end of a string of length i, $r+1 \leq i \leq n+1$, which contains $r+1$ 1's so that the last 1 is at the i-th position. For each i, there are $\binom{i-1}{r}$ strings. Summing over $i = r+1, \ldots, n+1$, we obtain the RHS.